Hariprasad Ravishankar

PRINCIPAL ENGINEER & SENIOR TEAM LEAD AT MATHWORKS

hariprasad.ravi@gmail.com github.com/hariprasadravi linkedin.com/in/hariprasadravi

MAJOR INITIATIVES

Compiler Optimization Framework (MLIR/CGIR)
Architected a modern, MLIR-based lowering pipeline, creating a unified IR layer that streamlines the transition from high-level frameworks (torch-mlir, tensorflow-mlir) to highly-optimized CUDA execution.

Hardware-Aware Optimization & SIMD Kernels
Designed and implemented vectorized and parallelized algorithms leveraging loop scheduling, cache-blocking, and register management for ARM (NEON) and x86 (AVX-512) architectures.

High-Level AI Primitive Design (dlarray)
Designed the end-to-end code generation for MATLAB's `dlarray` type, defining static/dynamic memory behavior and supporting 100+ operations. Implemented advanced permute optimizations to minimize data movement and maximize throughput.

Product Creation: AI Framework Interoperability
Spearheaded the design and launch of a new product enabling direct C++/CUDA code generation from PyTorch and TFLite models, culminating in the official MATLAB Coder Support Package for R2026a.

Model Compression & Specialized Data Types
Developed the `taylorPrunableNetwork` algorithm for network pruning and implemented BF16 compression solutions. Spearheaded end-to-end code generation for quantized and low-precision models, significantly reducing memory bandwidth and latency.

WORK EXPERIENCE

MathWorks, Principal Engineer & Senior Team Lead

May 2026 - Present

- Guide a high-performing compiler team towards advanced AI compilation solutions, focusing on MLIR/CGIR pipeline optimizations and SIMD/CUDA kernel efficiency.
- Launched the **MATLAB Coder Support Package for PyTorch and LiteRT Models**, enabling direct, production-grade C++/CUDA code generation from PyTorch and TFLite models.
- Maintainer of **Predict**, **Stateful Predict**, **Stateful Classify**, and **Classify** blocks in the Deep Learning Toolbox's Simulink library, supporting robust deployment pipelines.
- Optimize deep learning compiler code generation and execution utilizing CUDA, cuBLAS, cuDNN, and TensorRT to maximize hardware utilization.

MathWorks, Senior Team Lead & Senior Software Engineer

2019 - May 2026

- Led a team of 3 high-performing software engineers managing technical execution, mentoring, and professional development.
- Defined and drove the product roadmap for deep learning compiler technologies, presenting strategy and progress to **Vice Presidents and executive leadership** bi-annually.
- Architected end-to-end C/C++/CUDA code generation from PyTorch, TFLite, and MATLAB frameworks.
- Drove the technical integration of deep learning code generation into **Simulink**, supporting the Predict, LiteRT, and PyTorch blocks with full compatibility for **Rapid Acceleration**, **Model Reference Acceleration**, and advanced cache handling.
- Optimized compiler-generated code for deep learning models using CUDA, cuBLAS, cuDNN, and TensorRT libraries.

  Key Projects:
  - **MATLAB Coder Support Package for PyTorch and LiteRT Models (R2026a): Led the development and successful launch of this package, enabling seamless code generation workflows across MATLAB Coder, GPU Coder, Simulink, and Simulink Coder.**
  - **Code generation support for SSD object detection using cuDNN and TensorRT library.**
  - **Compiler optimizations to fuse convolution, scaling, zero-padding layers.**
  - **Code generation support for RNN, multi-input, multi-output networks, GANs, attention models.**

MathWorks, Software Engineer in Test

2018 - 2019

- Responsible for enhancing testability, developing test suites, and conducting hands-on testing for GPU code-generation and deep learning deployment capabilities.
- Tested deployment of Deep Learning networks to Embedded targets like GPUs, FPGAs, and ARM processors.
- Tested code-generation engine from MATLAB & Simulink, code-generation optimizations, and interfacing generated code with hardware targets.

  Key Projects:
  - Added code generation support for `crop2dLayer` and **Fully Convolutional Networks** for Semantic Segmentation (R2019a).
  - Designed and implemented layers to support code generation for **YOLO V2** for cuDNN and TensorRT (R2019a).
  - Added code generation support for **GoogleNet** and **Inception V3** for TensorRT target (R2019a).

MathWorks, Application Support Engineer

2017 - 2018

- Technical specialist in MATLAB and MATLAB products in Neural Networks, Image Processing, Deep Learning, Machine Learning, and Optimization algorithms.
- Solved complex technical issues for users in academia and industry, collaborating with development.
- Supported development in identifying quality issues, testing performance enhancements, and improving customer deep learning workflows.

  Key Projects:
  - Implemented MaxUnpooling Layer in C++-cuDNN and added support for **SegNet** within the GPU Coder Toolbox in MATLAB.
  - Added support for code generation for Denoising Networks such as **DnCNN** in the GPU Coder Toolbox.

NinePoint Medical, Deep Learning Co-op

2017

- Developed Deep Learning Architecture for efficient segmentation of clinical features in OCT images using TensorFlow.
- Designed custom networks based on well-known architectures such as U-Net and Fully Convolutional Net for image segmentation.
- Developed verification scripts and post-processing algorithms for offline deployment.
- Feature pushed to production with official **FDA clearance**.

Philips Research, Summer Research Intern

Jun 2016 - Jul 2016

- Worked on a Deep Learning Approach for Nucleus Segmentation in Cervical Cancer Images.
- Implemented *Segmentation by Classification* and *Segmentation using Deconvolution* techniques using DIGITS by NVIDIA.
- Designed and trained an architecture based on AlexNet and a 2-stage Deconvolution network for segmentation.
- Detected and segmented unseen nuclei with a PPV of 91% and Jaccard Score of 95%, making it viable for commercial deployment.

EDUCATION

MS in Electrical Engineering	, University of Southern California	**GPA 3.86/4.0**	**B.Tech in Electronics Engineering**	, Vellore Institute of Technology	**CGPA 9.13/10**

SKILLS SUMMARY

Expertise: C++, MATLAB, Simulink, CUDA, Python, PyTorch, LiteRT/TFLite, MLIR, CGIR
Libraries/APIs: TensorRT, cuDNN, cuBLAS, OpenCV
Leadership: Team Management, Roadmap Planning, Executive Presentations